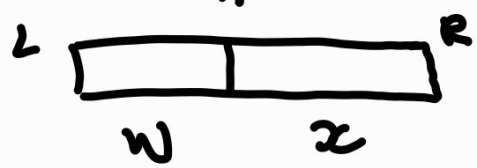
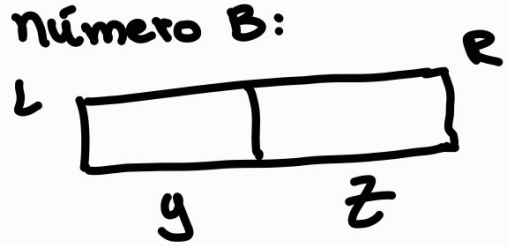
número A:



L -> esquerda
R -> direita

número B:



$$AB = (10^{n/2} \cdot w + x)(10^{n/2} \cdot y + z) \Rightarrow$$

$$AB = 10^{n} wy + 10^{n/2} wz + 10^{n/2} xy + xz \Rightarrow$$

$$AB = 10^{n} wy + 10^{n/2}(wz + xy) + xz \rightarrow \text{resultado obtido por dedução}$$

$$AB = 10^{n} wy + \underbrace{10^{n/2}(wz + xy)} + xz$$

♥ ideia de Gauss para multiplicação de números complexos

$$P_1 = wy$$

$$P_2 = xz$$

$$* P_3 = (w + x)(z + y)$$

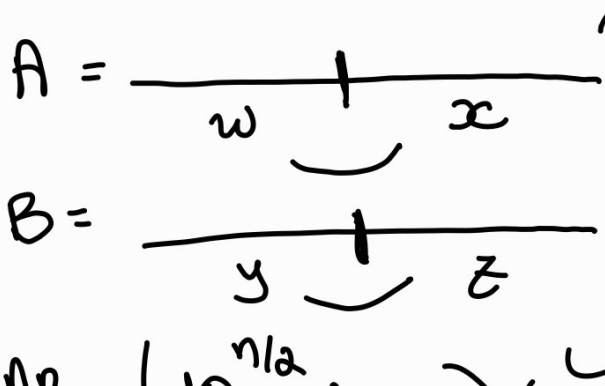
$$* P_3 = (w+x)(z+y) \Rightarrow wz + \underbrace{wy}_{P_1} + \underbrace{xz}_{P_2} + xy \Rightarrow$$

$$wz + P_1 + P_2 + xy \Rightarrow wz + xy = P_3 - P_1 - P_2$$

$$AB = 10^{n} P_1 + (P_3 - P_1 - P_2) + P_2$$

dado dois números: A e B

binário ↓ $10 \to 2$



decimais

$n = \max(|A|, |B|)$

$AB = (10^{n/2} \cdot w + x)(10^{n/2} \cdot y + z) =$

$= 10^{n/2} w\, 10^{n/2} \cdot y + 10^{n/2} wz + x \cdot 10^{n/2} \cdot y + xz =$

$= 10^{n} wy + 10^{n/2}(wz + xy) + xz$

- $P_1 = wy$
- $P_2 = xz$
- $P_3 = (w+x)(y+z)$

$\alpha \to P_3 = \underbrace{wy}_{P_1} + wz + xy + \underbrace{xz}_{P_2}$

$\alpha = P_3 - P_1 - P_2$

$\hookrightarrow AB = 10^{n} P_1 + 10^{n/2}(P_3 - P_1 - P_2) + P_2$